Sensor Fault Detection

One sensor rule. Two possible diagnoses.

One quantum oracle query per execution.

CONSTANT → FAULTY

Both inputs produce the same output.

BALANCED → WORKING

The two inputs produce different outputs.

Deutsch’s Algorithm | Qiskit + Aer | Colab-compatible notebook

1. Problem statement (PS)

The task

A single-bit sensor follows a hidden rule f.

Input: 0 or 1. Output: 0 or 1.

Determine whether f is constant or balanced.

Classify each rule with one quantum query.

Decision rule

$f(0) = f(1) \rightarrow$ Faulty

$f(0) \neq f(1) \rightarrow$ Working

Scope: a simulated sensor with coherent oracle access; no physical sensor is connected.

2. All four possible sensor rules

Hidden rule	f(0)	f(1)	Class	Diagnosis
Always 0	0	0	Constant	Faulty
Always 1	1	1	Constant	Faulty
Return input x	0	1	Balanced	Working
Invert input 1 XOR x	1	0	Balanced	Working

Required answer: the class of the function, not its exact identity.

3. Why use Deutsch's algorithm?

CLASSICAL EXACT METHOD

Evaluate f(0).

Evaluate f(1).

Compare the two outputs.

Worst case: 2 oracle queries.

DEUTSCH'S METHOD

Prepare a quantum state.

Query the oracle coherently.

Interfere and measure the input.

Ideal: 1 oracle query.

The advantage concerns oracle calls; simulator runtime is not the benchmark.

4. Methodology: from model to evidence

01

MODEL

Define all four rules.

02

BUILD

Construct reversible oracles.

03

VALIDATE

Check truth mappings and unitarity.

04

CLASSIFY

Run one-query Deutsch circuits.

05

ANALYSE

Compare theory, counts and noise.

Reproducible experiment: fixed seeds, recorded versions, assertions and saved outputs.

5. Architecture: two qubits, one measured output

PREPARE

q0 = |0⟩

q1 = |1⟩

SUPERPOSE

H on q0

H on q1

ORACLE

One call

to U_f

INTERFERE

H on q0

only

MEASURE

q0 → c0

0 or 1

RESULT 0

Constant → faulty

RESULT 1

Balanced → working

6. How the algorithm works

SUPERPOSITION

H prepares the input in the plus state and the helper in the minus state.

PHASE KICKBACK

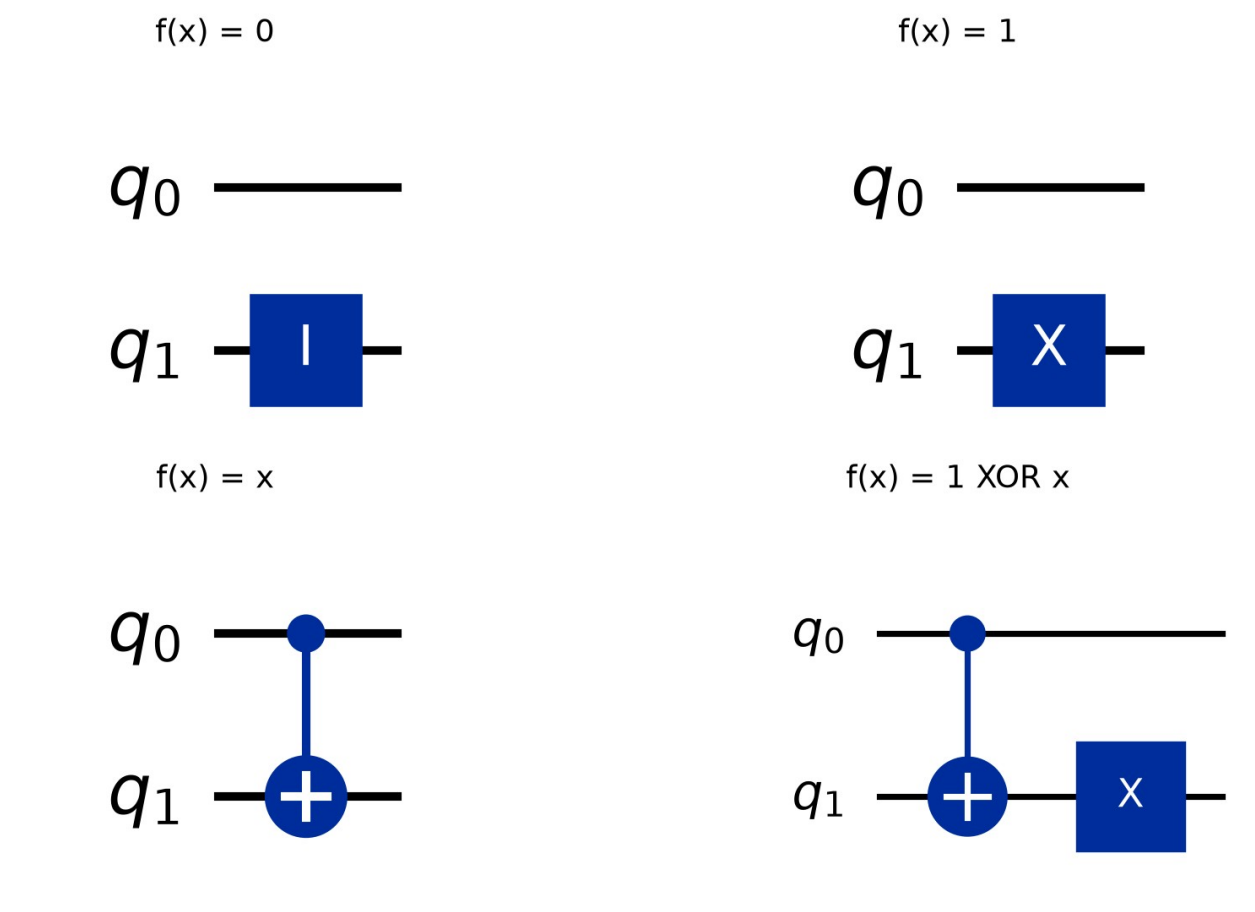
The oracle creates equal or opposite signs on the input amplitudes.

INTERFERENCE

The final H turns the sign difference into a definite 0 or 1.

Measured result = f(0) XOR f(1). Equal outputs → 0; different outputs → 1.

7. Solution: build every oracle



ORACLE CONTRACT

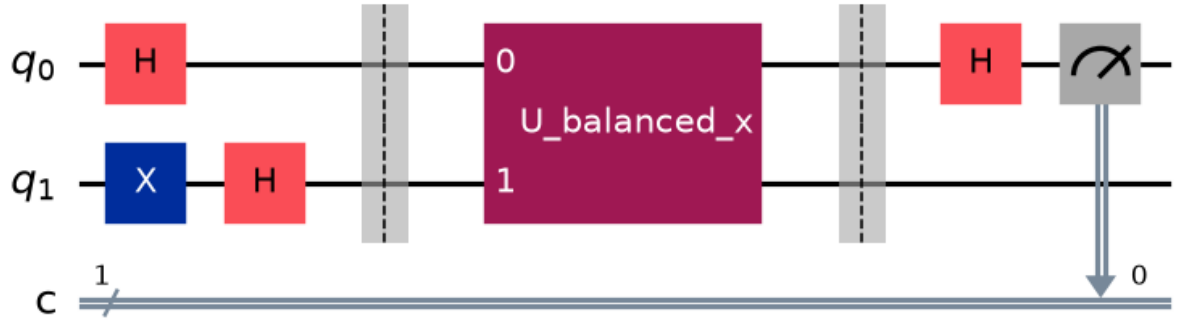
Keep input x unchanged.

Change helper y to:

y XOR f(x).

This preserves reversibility.

8. Implemented circuit and simulation



QISKIT

Construct circuits and transpile.

AER SIMULATOR

2 qubits; 1 classical bit.

IDEAL EXPERIMENT

1,024 shots per oracle; seed 42.

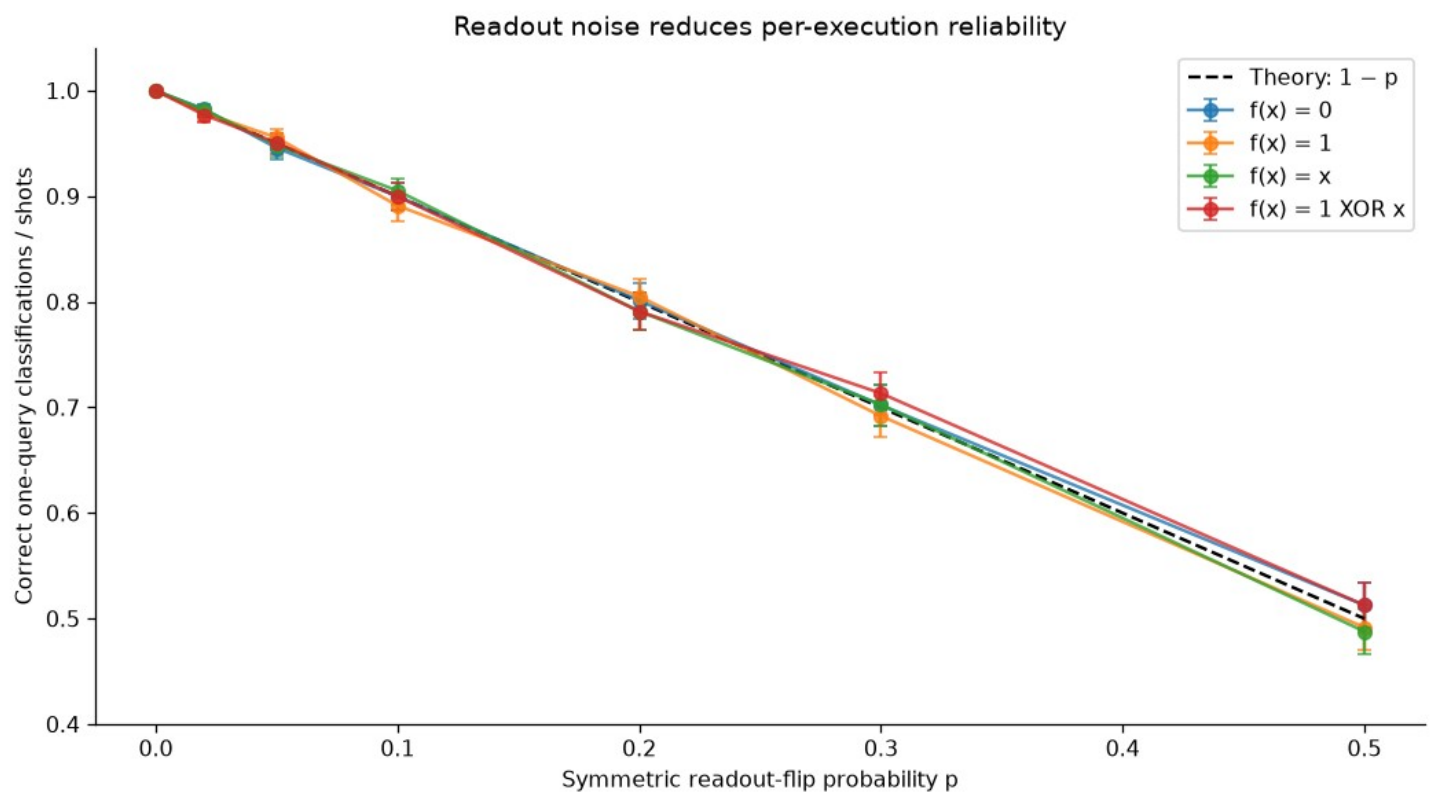
9. Results: all four diagnoses are correct

Rule	Count 0	Count 1	Measured bit	Diagnosis
0	1024	0	0	Faulty
1	1024	0	0	Faulty
x	0	1024	1	Working
1 XOR x	0	1024	1	Working

All ideal classifications matched the exact statevector predictions.

One query per shot × 1,024 shots × 4 oracles = 4,096 total ideal queries.

10. Validation and measurement-noise study



CORRECTNESS CHECKS

16 basis mappings passed.

4 unitarity checks passed.

NOISE FINDING

Expected accuracy = 1 − p.

Only measurement errors are modelled.

11. What the project demonstrates

ACHIEVED

All four oracles implemented.

One-query ideal classification.

Theory agrees with simulation.

Noise effects visualised.

LIMITATIONS

Simulated coherent sensor model.

No physical sensor or hardware run.

No gate-noise or decoherence model.

No practical runtime-speedup claim.

Next step: test calibrated device noise and account for all repeated queries.

12. Closing summary and questions

"One coherent query reveals whether the sensor rule is constant or balanced."

0 → FAULTY

Constant rule

1 → WORKING

Balanced rule

Project: github.com/Jeshurun-Pon-Mcgrath/Jeshurun_QC

References: Deutsch (1985), doi:10.1098/rspa.1985.0070; IBM Quantum learning; Qiskit QuantumCircuit documentation; Qiskit Aer ReadoutError documentation.